DoubleMLDeep: Estimation of Causal Effects with LLMs and Multimodal Data

Ausschuss für Ökonometrie

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Motivation

Motivation

	A	B	C	D	E	F	G	H	I	J	K
1	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	Y
2	0,26222194	0,73438599	0,44667203	0,97138377	0,081064	0,28441456	0,30204598	0,42916989	0,45202869	0,48458485	0,69688241
3	0,02086748	0,76377164	0,82984213	0,44114913	0,6791843	0,10440879	0,06871173	0,65049866	0,38145381	0,37883627	0,68724919
4	0,80214042	0,60108851	0,3924631	0,64680441	0,52447716	0,71441139	0,92058036	0,50487157	0,0639613	0,72018174	0,70586344
5	0,69860016	0,48544746	0,02697306	0,56612716	0,64030572	0,38122836	0,77357736	0,7665354	0,56153968	0,40036439	0,96259876
6	0,76871281	0,52693951	0,23152259	0,99343286	0,19512117	0,52281687	0,74631891	0,51990814	0,8916446	0,13854693	0,15014547
7	0,80559982	0,49248903	0,52094313	0,83491804	0,19224862	0,51690107	0,99896405	0,21315395	0,13844972	0,5972433	0,30108879
8	0,14274622	0,68082916	0,21375762	0,12917741	0,101114	0,35962408	0,37101036	0,88481715	0,0917219	0,81144147	0,23748577
9	0,5565959	0,27805181	0,38191987	0,96746964	0,88262331	0,64204409	0,61391038	0,14798767	0,75991882	0,71364563	0,26831957
10	0,86680082	0,98940603	0,08032276	0,51284266	0,85067788	0,31879029	0,14212956	0,58889047	0,85003262	0,40984155	0,90320049
11	0,49294831	0,9930191	0,32376346	0,76486908	0,71150197	0,49307446	0,00732655	0,1366301	0,95413681	0,8192993	0,41659136
12	0,83367503	0,99203204	0,94500561	0,38442651	0,08267135	0,44319059	0,50597368	0,33789173	0,71150674	0,71723848	0,82498573
13	0,30901589	0,45442227	0,39182182	0,94058761	0,84662039	0,13237356	0,39203009	0,23863556	0,66904622	0,66739594	0,05703788
14	0,84854041	0,07220964	0,60564451	0,11235493	0,11423158	0,19425556	0,10525554	0,69400453	0,24273427	0,74521656	0,1934563
15	0,41204349	0,37890159	0,21172493	0,47214042	0,28972671	0,28045945	0,26144942	0,50182126	0,56848988	0,34171099	0,55046509
16	0,8270131	0,7550563	0,68080658	0,99718432	0,54748336	0,07330951	0,96389321	0,43492645	0,32444209	0,37929049	0,54724228
17	0,65968289	0,42809072	0,50651859	0,85136778	0,67725248	0,80776288	0,80154682	0,41961695	0,03794024	0,10550978	0,18250098
18	0,00250478	0,10976101	0,58665193	0,59539311	0,06932994	0,44532097	0,38697059	0,86107306	0,32445673	0,03897245	0,4593923
19	0,1646558	0,11563336	0,54131562	0,70212986	0,78935386	0,65759136	0,96083497	0,18585388	0,02245136	0,08388247	0,9684134

Motivation

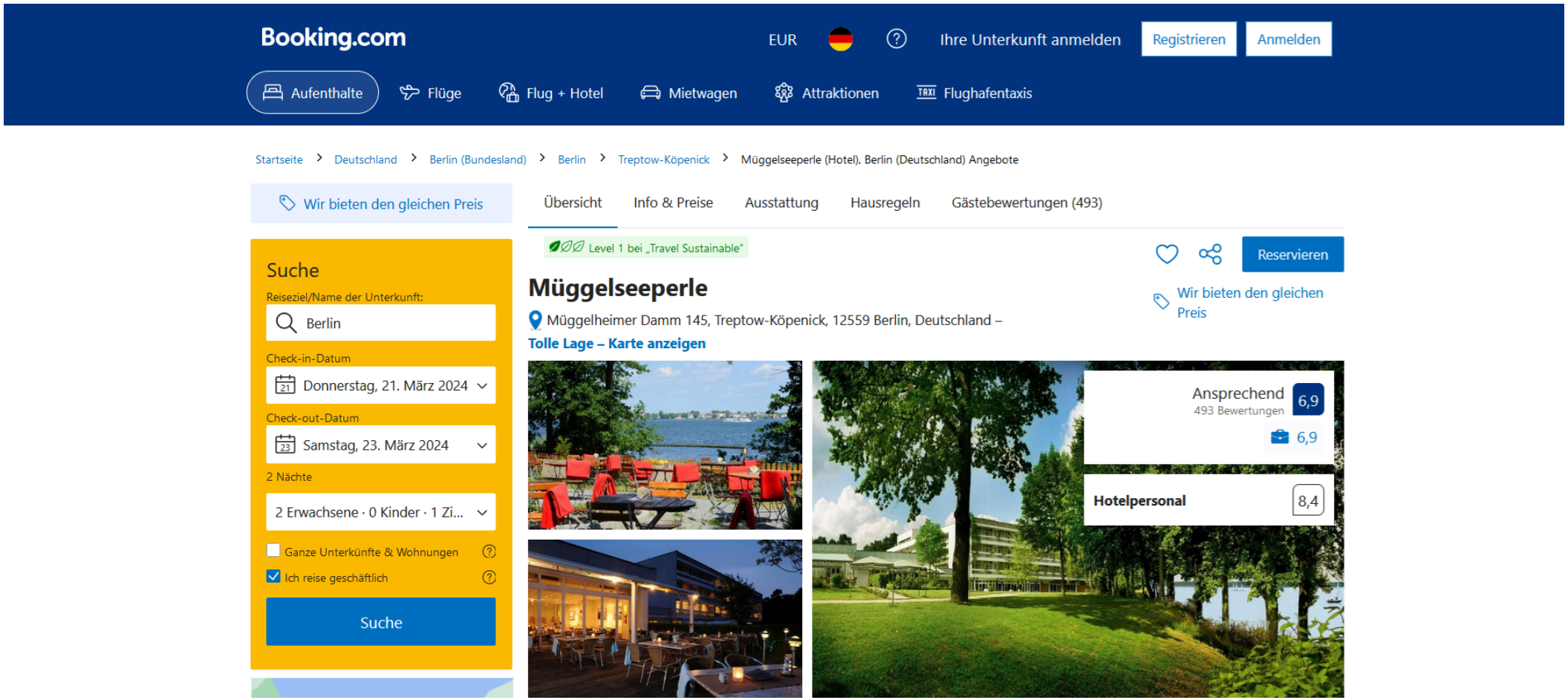


Motivation

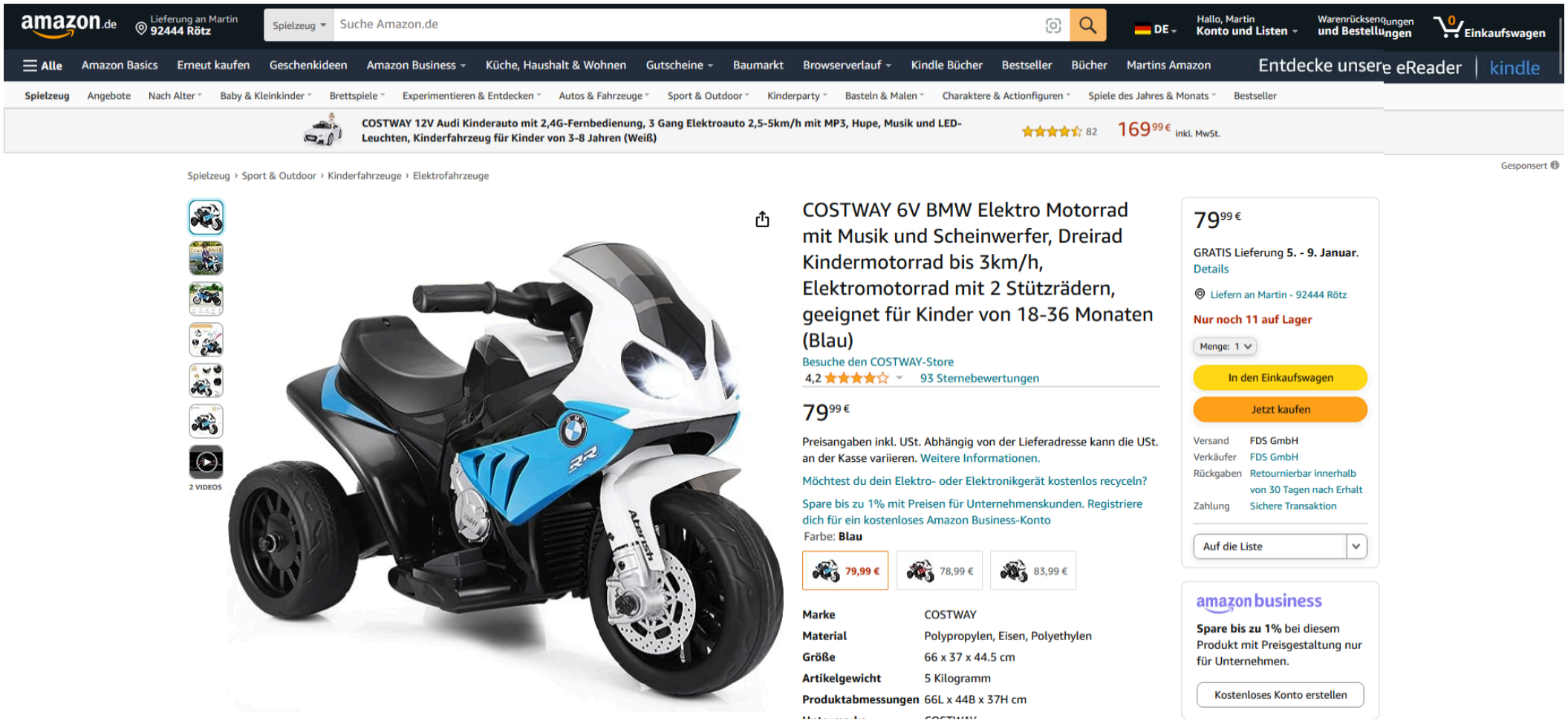




Motivation


Booking.com
EUR
Ihre Unterkunft anmelden
Registrieren
Anmelden
Aufenthalte
Flüge
Flug + Hotel
Mietwagen
Attraktionen
Flughafentaxis
Startseite › Deutschland › Berlin (Bundesland) › Berlin › Treptow-Köpenick › Müggelseeperle (Hotel), Berlin (Deutschland) Angebote
Wir bieten den gleichen Preis
Suche
Reiseziel/Name der Unterkunft:
Berlin
Check-in-Datum
Donnerstag, 21. März 2024
Check-out-Datum
Samstag, 23. März 2024
2 Nächte
2 Erwachsene · 0 Kinder · 1 Zi...
Ganze Unterkünfte & Wohnungen
Ich reise geschäftlich
Suche
Übersicht
Info & Preise
Ausstattung
Hausregeln
Gästebewertungen (493)
Level 1 bei „Travel Sustainable"
Reservieren
Müggelseeperle
Müggelheimer Damm 145, Treptow-Köpenick, 12559 Berlin, Deutschland –
Tolle Lage – Karte anzeigen
Wir bieten den gleichen Preis
Ansprechend
493 Bewertungen
6,9
6,9
Hotelpersonal
8,4

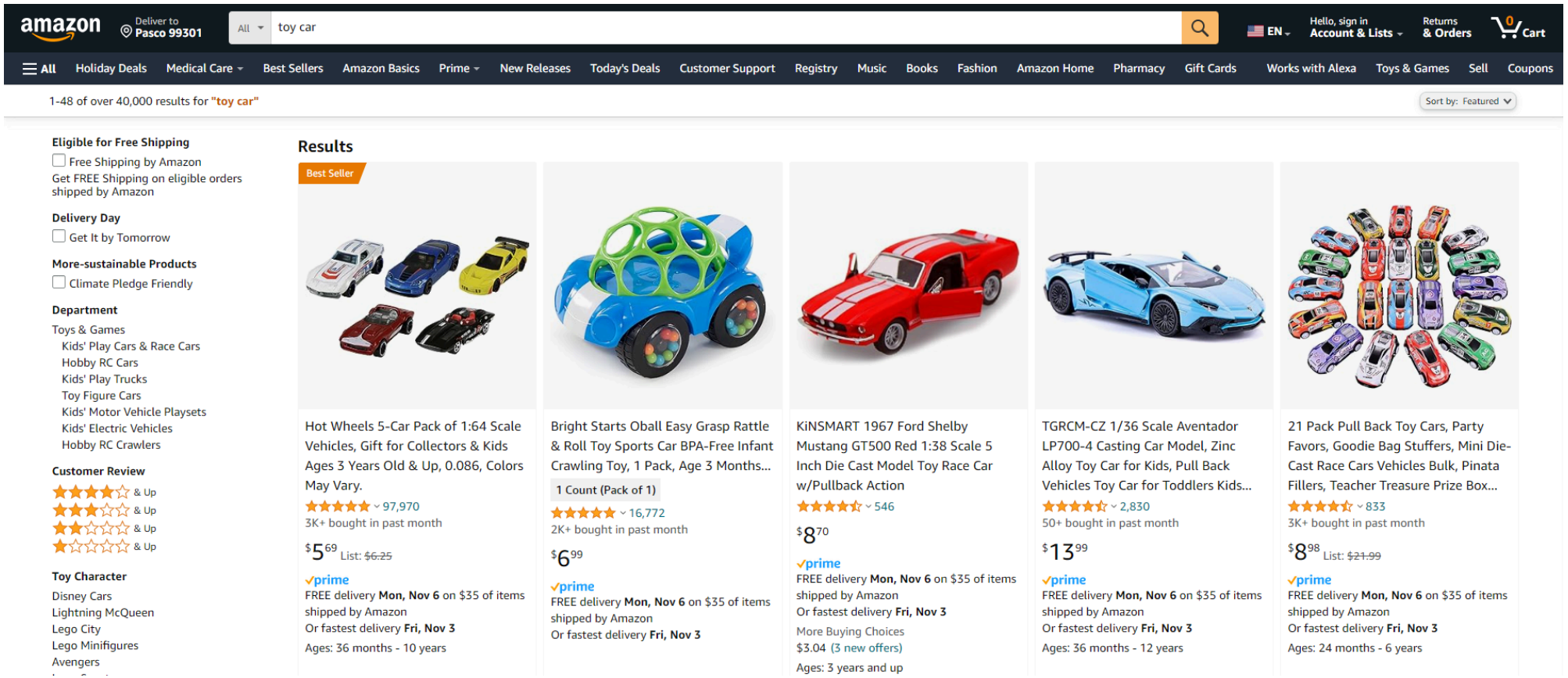
Motivation


amazon.de
Lieferung an Martin
92444 Rötz
Spielzeug
Suche Amazon.de
DE
Hallo, Martin
Konto und Listen
Warenrücksendungen
und Bestellungen
Einkaufswagen
Alle
Amazon Basics
Erneut kaufen
Geschenkideen
Amazon Business
Küche, Haushalt & Wohnen
Gutscheine
Baumarkt
Browserverlauf
Kindle Bücher
Bestseller
Bücher
Martins Amazon
Entdecke unsere eReader
kindle
Spielzeug
Angebote
Nach Alter
Baby & Kleinkinder
Brettspiele
Experimentieren & Entdecken
Autos & Fahrzeuge
Sport & Outdoor
Kinderparty
Basteln & Malen
Charaktere & Actionfiguren
Spiele des Jahres & Monats
Bestseller
COSTWAY 12V Audi Kinderauto mit 2,4G-Fernbedienung, 3 Gang Elektroauto 2,5-5km/h mit MP3, Hupe, Musik und LED-Leuchten, Kinderfahrzeug für Kinder von 3-8 Jahren (Weiß)
82
169 99 € inkl. MwSt.
Gesponsert
Spielzeug › Sport & Outdoor › Kinderfahrzeuge › Elektrofahrzeuge
2 VIDEOS
COSTWAY 6V BMW Elektro Motorrad mit Musik und Scheinwerfer, Dreirad Kindermotorrad bis 3km/h, Elektromotorrad mit 2 Stützrädern, geeignet für Kinder von 18-36 Monaten (Blau)
Besuche den COSTWAY-Store
4,2
93 Sternebewertungen
79 99 €
Preisangaben inkl. USt. Abhängig von der Lieferadresse kann die USt. an der Kasse variieren. Weitere Informationen.
Möchtest du dein Elektro- oder Elektronikgerät kostenlos recyceln?
Spare bis zu 1% mit Preisen für Unternehmenskunden. Registriere dich für ein kostenloses Amazon Business-Konto
Farbe: Blau
79,99 €
78,99 €
83,99 €
Marke COSTWAY
Material Polypropylen, Eisen, Polyethylen
Größe 66 x 37 x 44.5 cm
Artikelgewicht 5 Kilogramm
Produktabmessungen 66L x 44B x 37H cm
79 99 €
GRATIS Lieferung 5. - 9. Januar.
Details
Liefern an Martin - 92444 Rötz
Nur noch 11 auf Lager
Menge: 1
In den Einkaufswagen
Jetzt kaufen
Versand FDS GmbH
Verkäufer FDS GmbH
Rückgaben Retournierbar innerhalb von 30 Tagen nach Erhalt
Zahlung Sichere Transaktion
Auf die Liste
amazon business
Spare bis zu 1% bei diesem Produkt mit Preisgestaltung nur für Unternehmen.
Kostenloses Konto erstellen

Motivation

- Text and images contain important information for causal estimation.
- Here: Multimodal data as confounders
- Integrating of multimodal data into the Double / Debiased Machine Learning (DML)
- Applications in Marketing, Medicine / Health, Finance, ...
- Literature: Klaassen et al. (2024), Bach et al. (2023)

Application



Application: Estimation of Price Elasticity

- Understanding price elasticity is crucial for economic analysis and business decisions
  - It influences strategies, pricing, and market dynamics
- Traditional causal methods rely on structured data, limiting their applicability
- Utilizing unstructured data enhances decision-making
- Sources like images and text contain valuable insights
  - Contain rich hidden information about unobserved confounding variables

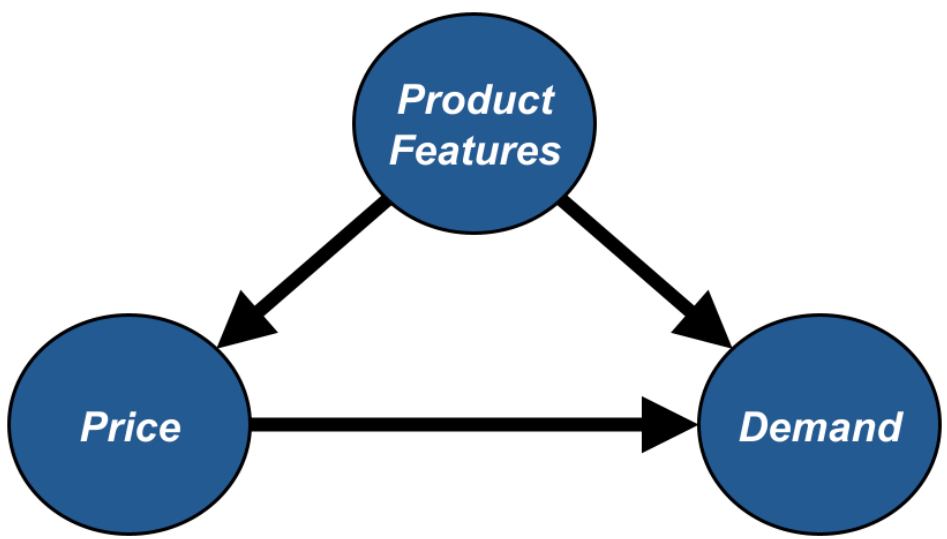
Double Machine Learning

Partially linear regression model (PLR)

$$
\begin{aligned}
Y &= D\theta_0 + g_0(X) + \varepsilon, & \mathbb{E}[\varepsilon|P, X] &= 0, \\
D &= m_0(X) + V, & \mathbb{E}[V|X] &= 0,
\end{aligned}
$$

with

- Outcome Y, e.g. Demand $\ln(Q)$
- Treatment variable D, e.g. Price $\ln(P)$
- Additional product features $X = (X_1, \ldots)$
- Stochastic errors ε and V



$\Rightarrow$ The causal parameter θ_0 can be interpreted as price elasticity of demand!

Double/Debiased Machine Learning (DML)

Double/Debiased Machine Learning (DML)

- Use machine learning methods to fit

$$l_0(X) := \mathbb{E}[Y||X]$$
$$m_0(X) := \mathbb{E}[D|X]$$

- Construct the orthogonalized score (either cross-fitting or split the sample) for a fitted nuisance learner $\hat{\eta}_0 := (\hat{l}_0, \hat{m}_0)$

$$\psi(W, \theta, \hat{\eta}_0) := \Big(Y - \hat{l}_0(X) - \theta\big(D - \hat{m}_0(X)\big)\Big)\Big(D - \hat{m}_0(X)\Big)$$

- Compute the estimate as the solution of

$$0 = \frac{1}{n}\sum_{i=1}^{n} \psi(W, \hat{\theta}_0, \hat{\eta}_0)$$

Double/Debiased Machine Learning (DML)

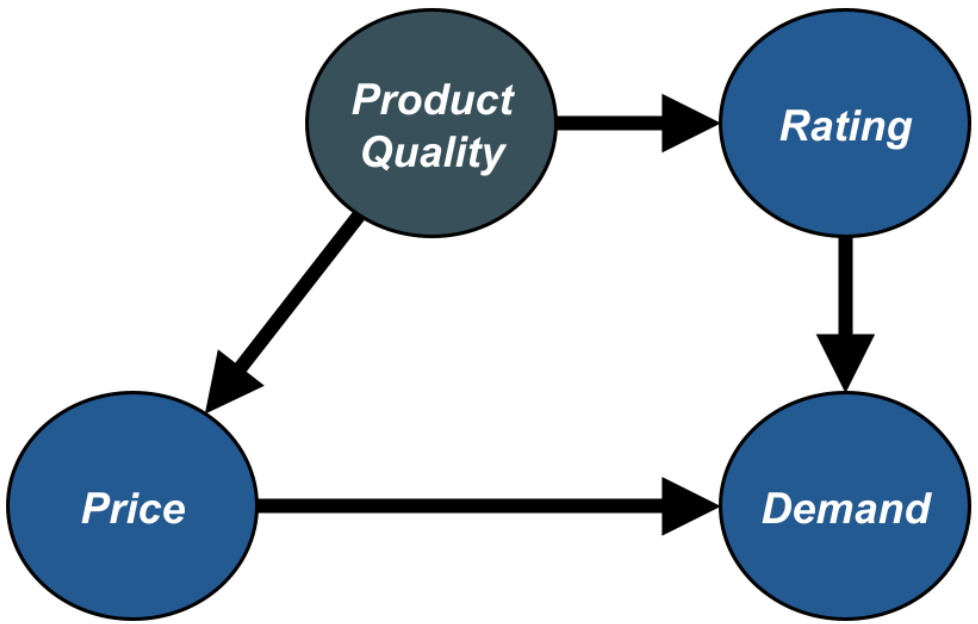
- Assumptions for the nuisance elements (see Chernozhukov et al. (2018))
  $\|\hat{m}_0 - m_0\|_{P,2} \times \left(\|\hat{m}_0 - m_0\|_{P,2} + \|\hat{\ell}_0 - \ell_0\|_{P,2}\right) \leq \delta_N N^{-1/2}$
- Under some regularity conditions the estimator $\hat{\theta}_0$ concentrates in $1/\sqrt{n}$-neighborhood of θ_0 and

$$\sqrt{n}(\hat{\theta}_0 - \theta_0) \to \mathcal{N}(0, \sigma^2)$$

$\Rightarrow$ simple approach to control for product features, *if they can be used for ML models!*

Application

- A presumably very important feature to consider might be product quality. How to determine product quality? Maybe use average rating, number of reviews, etc.

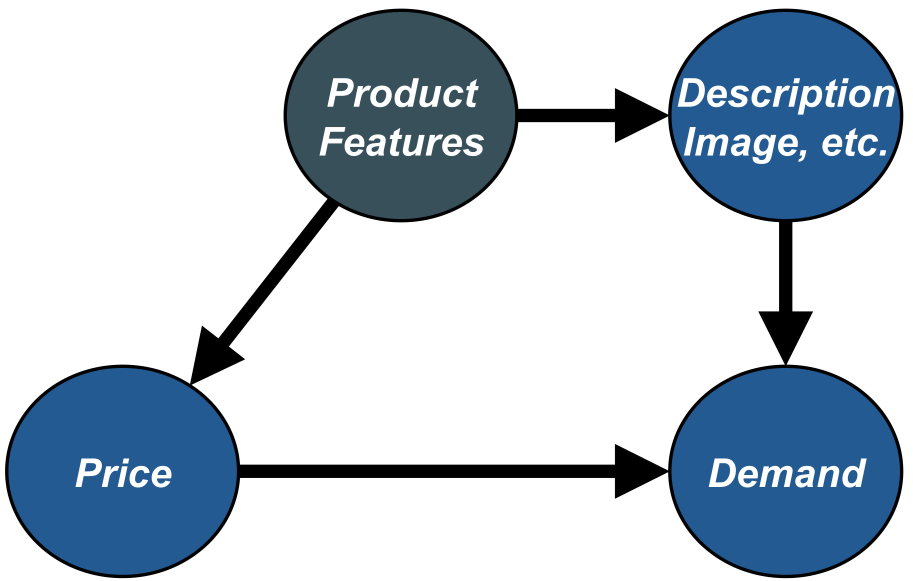


$\Rightarrow$ *Additionally rely on product description and images to describe the product feature!*

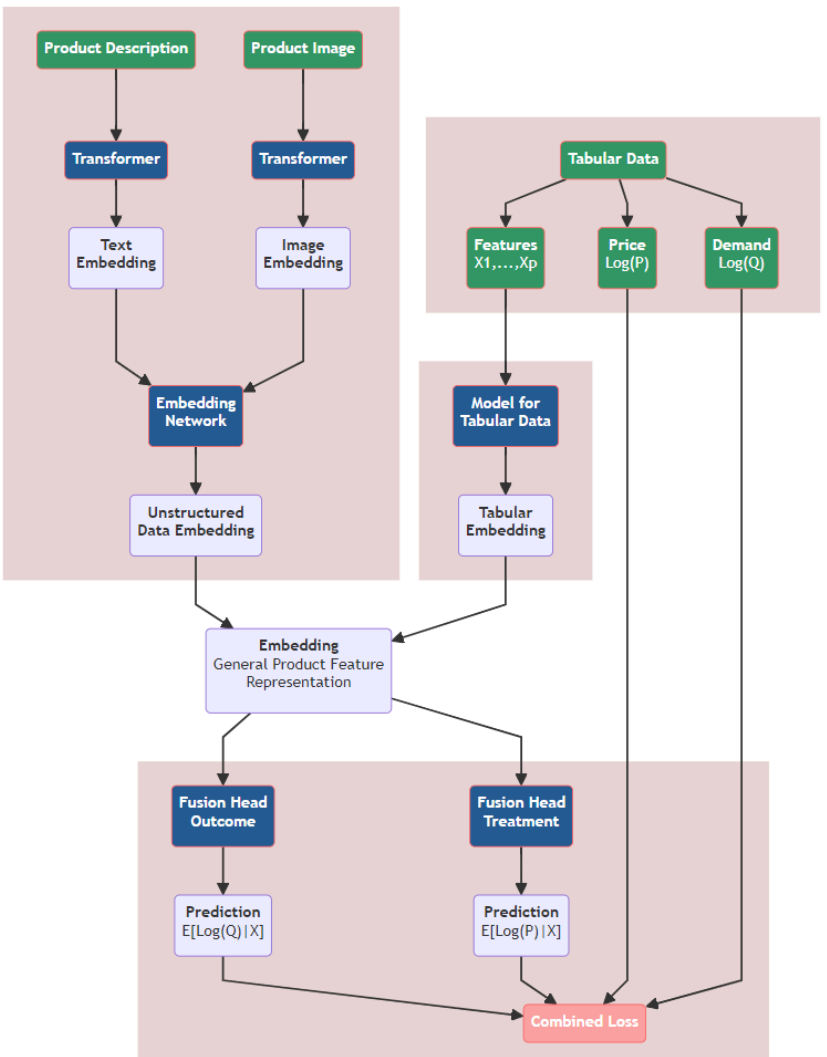
DoubleML Deep

DoubleML Deep - Motivation

- Use product description and image *additionally* to conventional tabular features

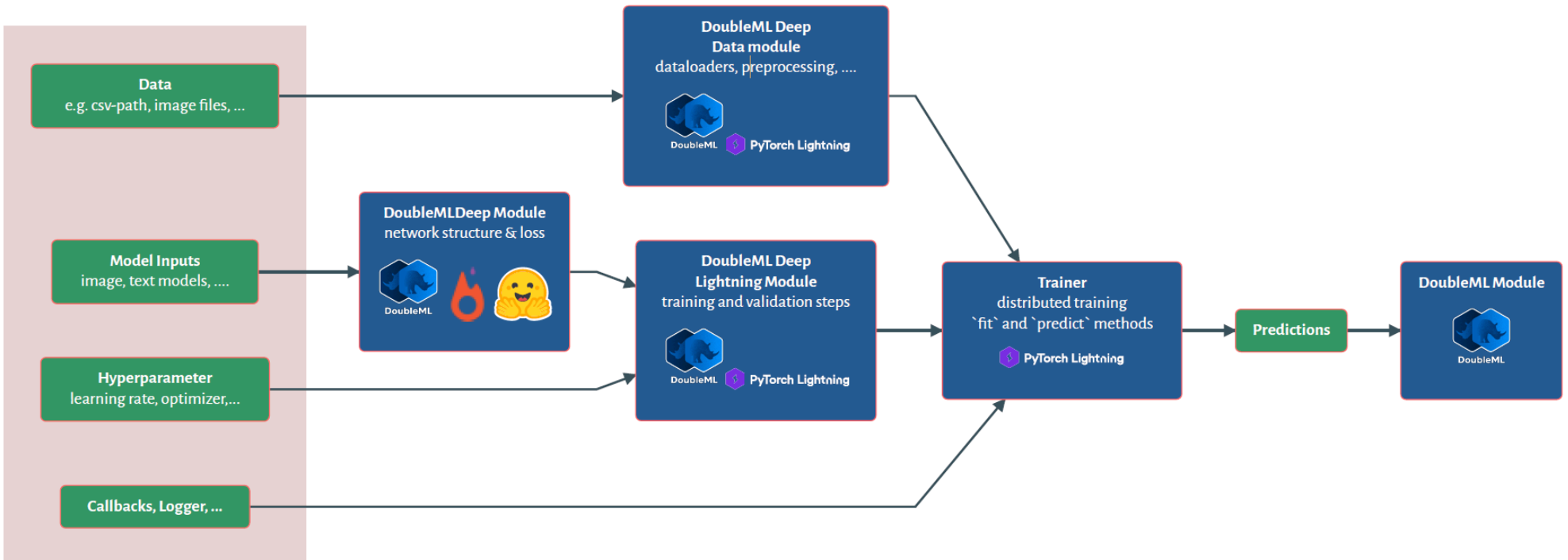


DoubleML Deep - Module Structure



High-Level PLR Model Architecture. Both nuisance components are trained simultaneously with a combined loss.

DoubleML Deep - Workflow



High-Level Workflow of *DoubleMLDeep*

⇒ More details can be found in Klaassen et al. (2024)

DoubleML Deep - More Details

- Use pretrained models for unstructured data e.g. Bert and Beit
- Basic models for tabular data seem sufficient
- Monitor the nuisance losses
  - $\|\hat{l}_0 - \ln(Q)\|_{P,2} \leq \|\hat{l}_0 - l_0\|_{P,2} + \sigma_{\tilde{\epsilon}}$
  - $\|\hat{m}_0 - \ln(P)\|_{P,2} \leq \|\hat{m}_0 - m_0\|_{P,2} + \sigma_V$

Application

Amazon Toys Dataset

- Public Data from the Amazon sales platform from category toys and games for subcategories vehicles, cars and trucks.
- Using the Sales Rank as a proxy for quantities, as shown in Bajari et al. (2023).

Variable	Description
Sales Rank	Sales rank as weighted mean for the last 30 days
Price	Price as weighted mean for the last 30 days
Text	Combination of Title, Category, Description, etc.
Image	First Image of the product

Amazon Toys Dataset

Continous Variables

- Reviews: Rating
- Reviews: Review Count
- New Offer Count: Current
- Count of retrieved live offers: New, FBA
- Count of retrieved live offers: New, FBM

Categorical Variables

- Lightning Deals: Upcoming Deal
- Buy Box: Is FBA

Amazon Toys Dataset

Data Examples

Kinsmart Set of 4 McLaren 720s Toy | ...



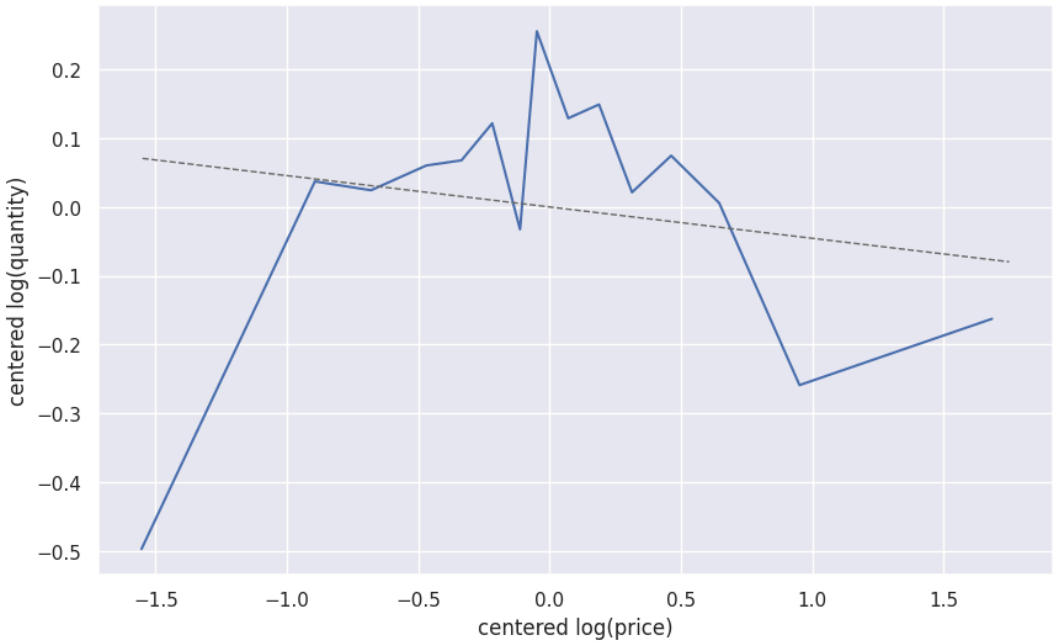
Pixar Cars Mack Uncle Lightning...



Baseline

- Baseline estimate with tabular covariates, OLS model for regression
  - $\ln(Q) = \theta_0 \ln(P) + \beta^T X + \epsilon$
    - $R^2 = 0.330$

2.5 %	θ	97.5 %
-0.072	-0.046	-0.019

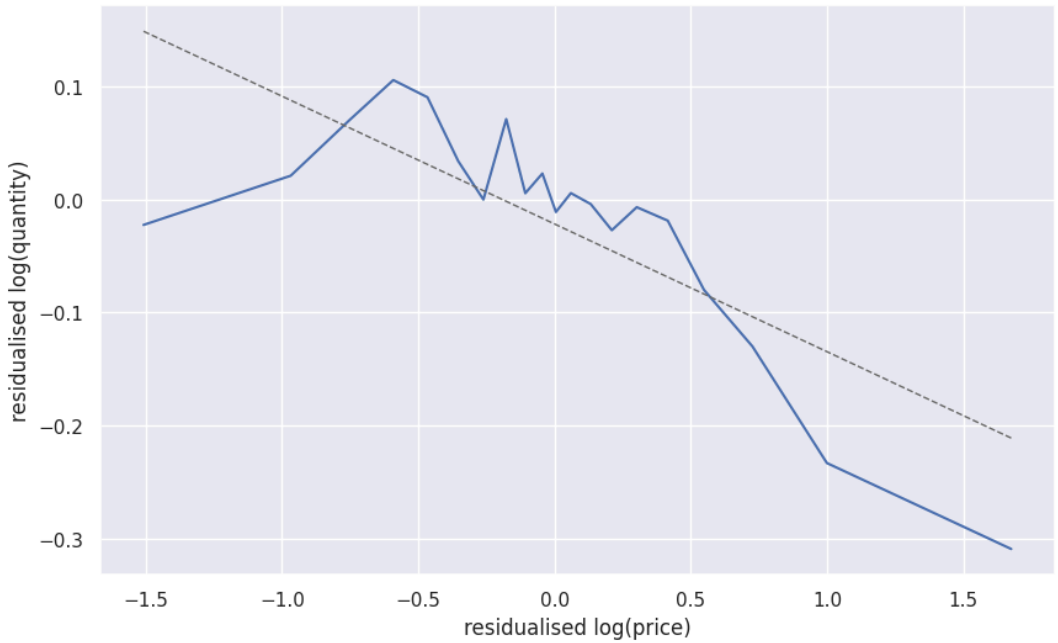


Centered logarithmic negative sales rank over centered logarithmic price as binned plot

Baseline

- Estimate with tabular data as covariates X in `DoubleMLPLR` model with `RandomForest` regressors
  - $l_0(X) := \mathbb{E}[\ln(Q)|X]$
    - $R^2_{l_0} = 0.5986$
  - $m_0(X) := \mathbb{E}[\ln(P)|X]$
    - $R^2_{m_0} = 0.1884$

2.5 %	θ	**97.5 %**
-0.132	-0.1098	-0.080



Residualised logarithmic negative sales rank over residualised logarithmic price as binned plot

Deep Learning Models

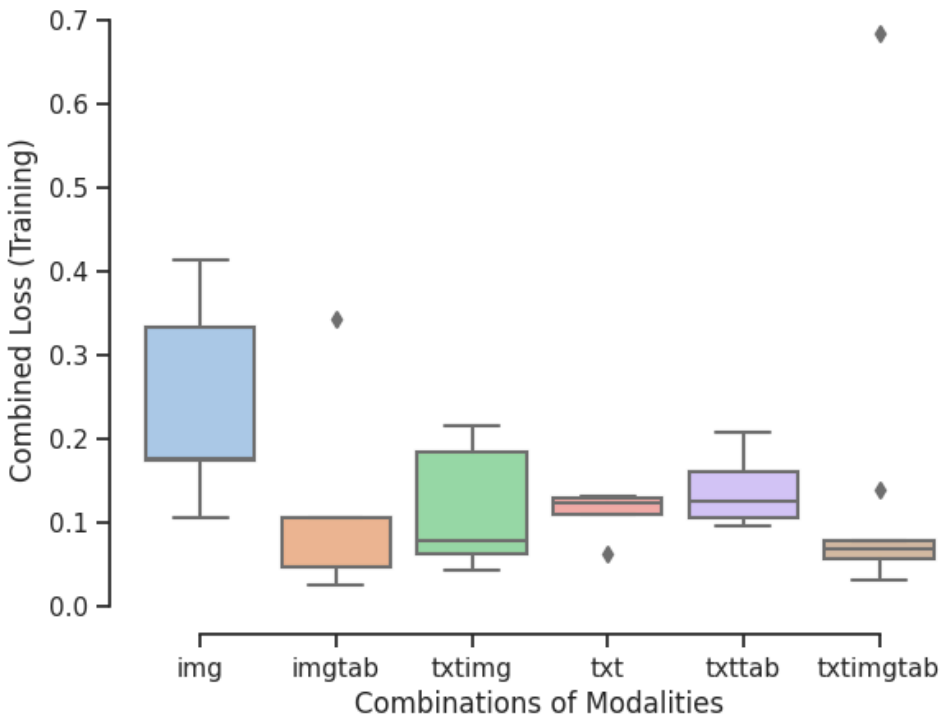
- In this study, a variety of input data combinations were used to train `DoubleMLPLRDeep` models [1]:

Key	Used Confounders
`img`	$X = (X_{img})$
`imgtab`	$X = (X_{img}, X_{tab})$
`txtimg`	$X = (X_{txt}, X_{img})$
`txt`	$X = (X_{txt})$
`txttab`	$X = (X_{txt}, X_{tab})$
`txtimgtab`	$X = (X_{txt}, X_{img}, X_{tab})$

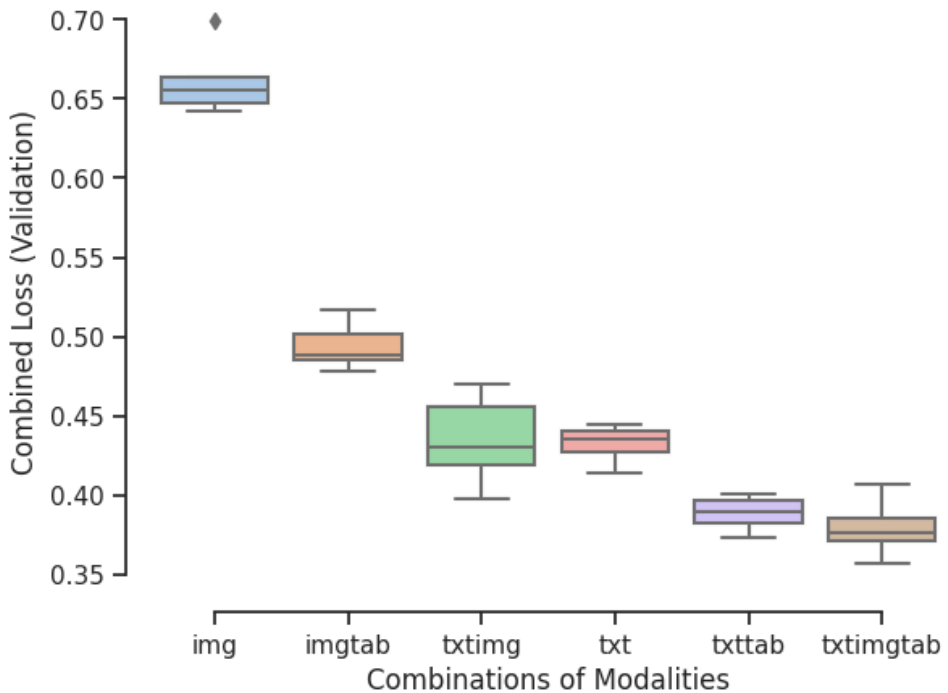
$\Rightarrow$ The aim is to emphasize the benefits of utilizing unstructured data.

Deep Learning Models: RMSE-Scores

- Combined RMSE Score (Training)

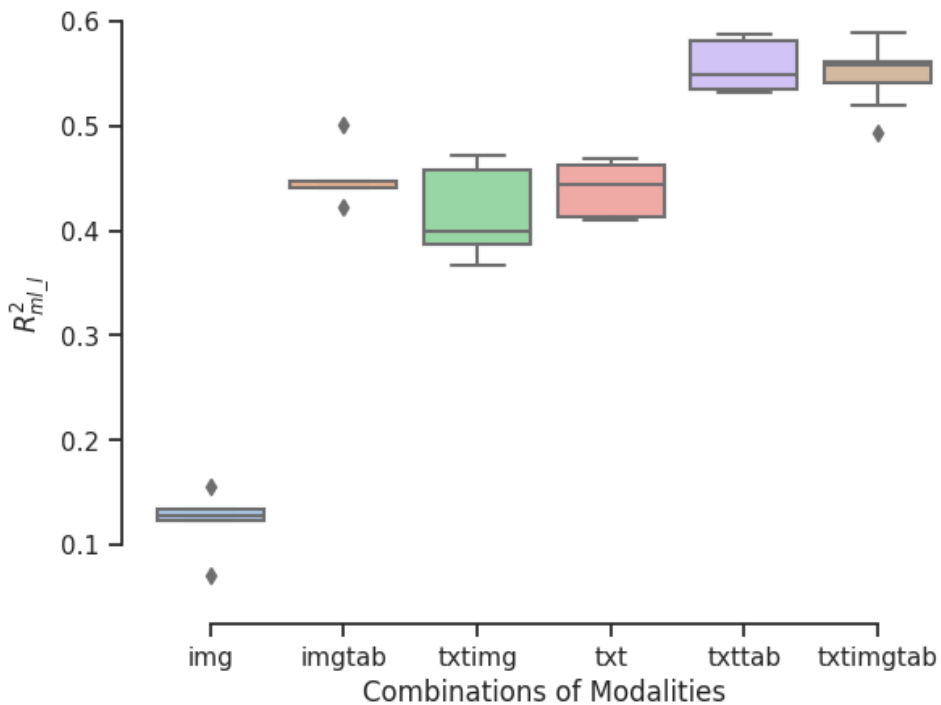


- Combined RMSE Score (Validation)

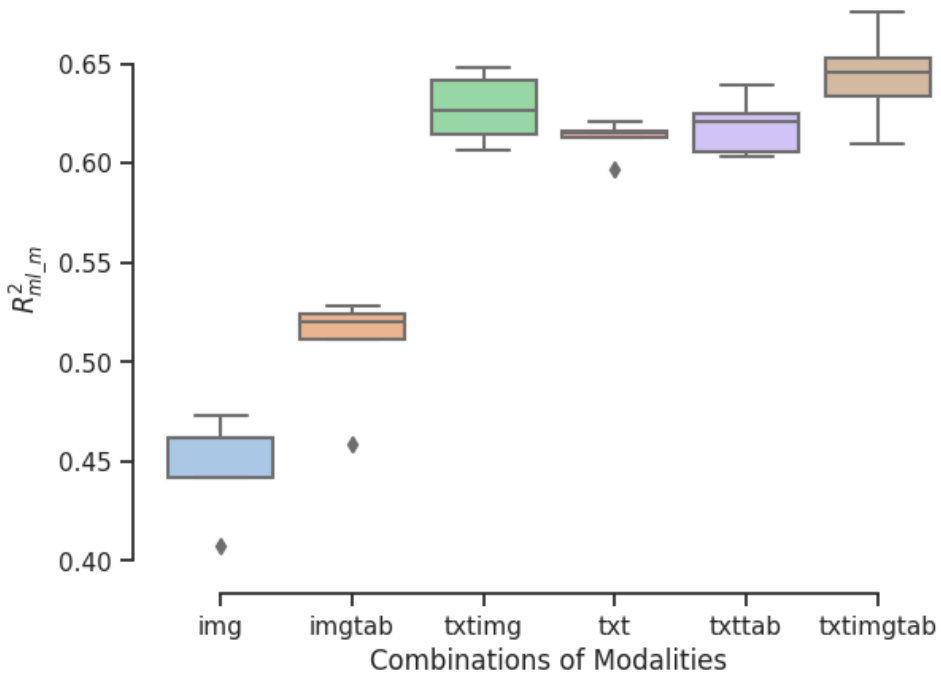


Deep Learning Models: R^2-Scores

- R^2 of log(Quantity) on Validation Set

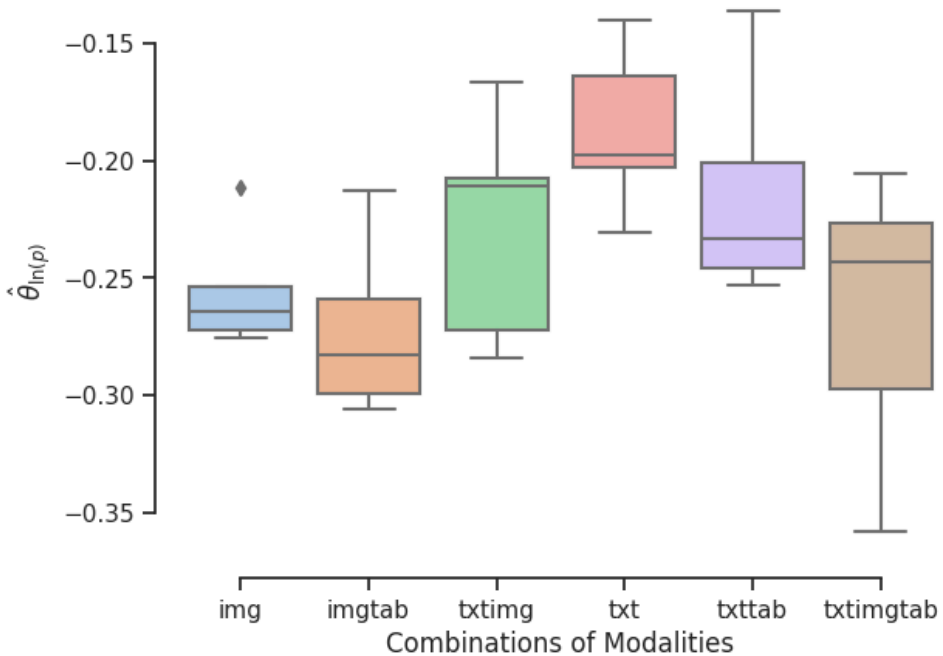


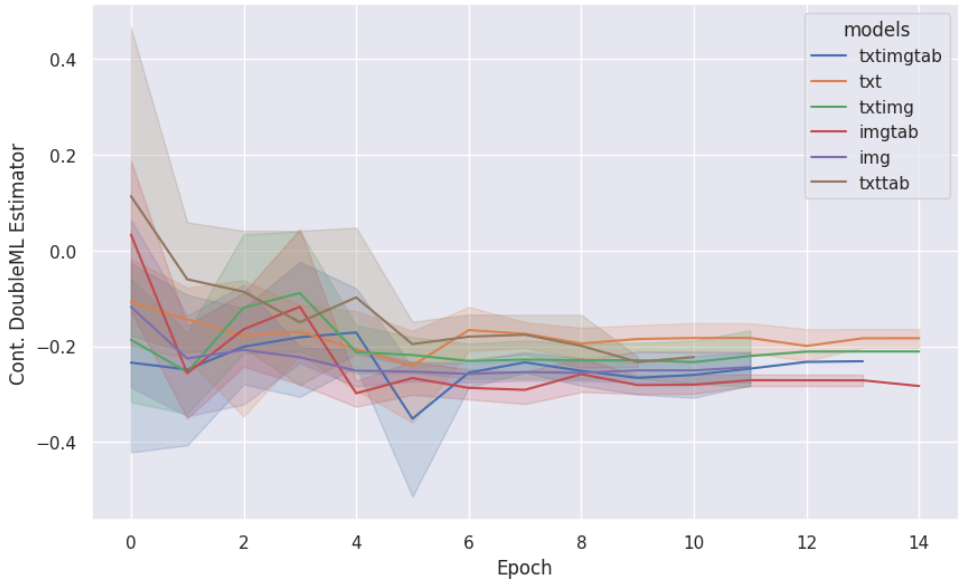
- R^2 of log(Price) on Validation Set



(First) Results

- Using different combinations of tabular, image and text data we obtain the following estimates



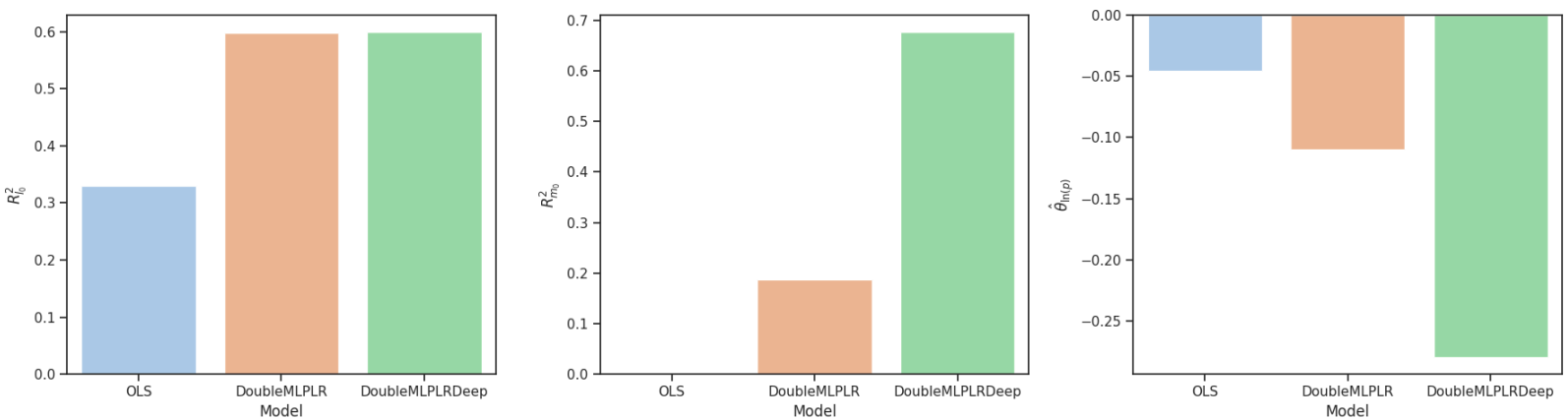


Continous Estimates from `DoubleMLPLR` on Validation Set during the Training Process

Comparison to Baseline Estimates

Model	Covariates	$R^2_{l_0}$	$R^2_{m_0}$	$\hat{\theta}_0$
`OLS`	$X = X_{tab}$	0.3300	-	-0.0455
`DoubleMLPLR`	$X = X_{tab}$	0.5986	0.1884	-0.1098
`DoubleMLPLRDeep`	$X = (X_{tab}, X_{txt}, X_{img})$	0.5990	0.6765	-0.2794

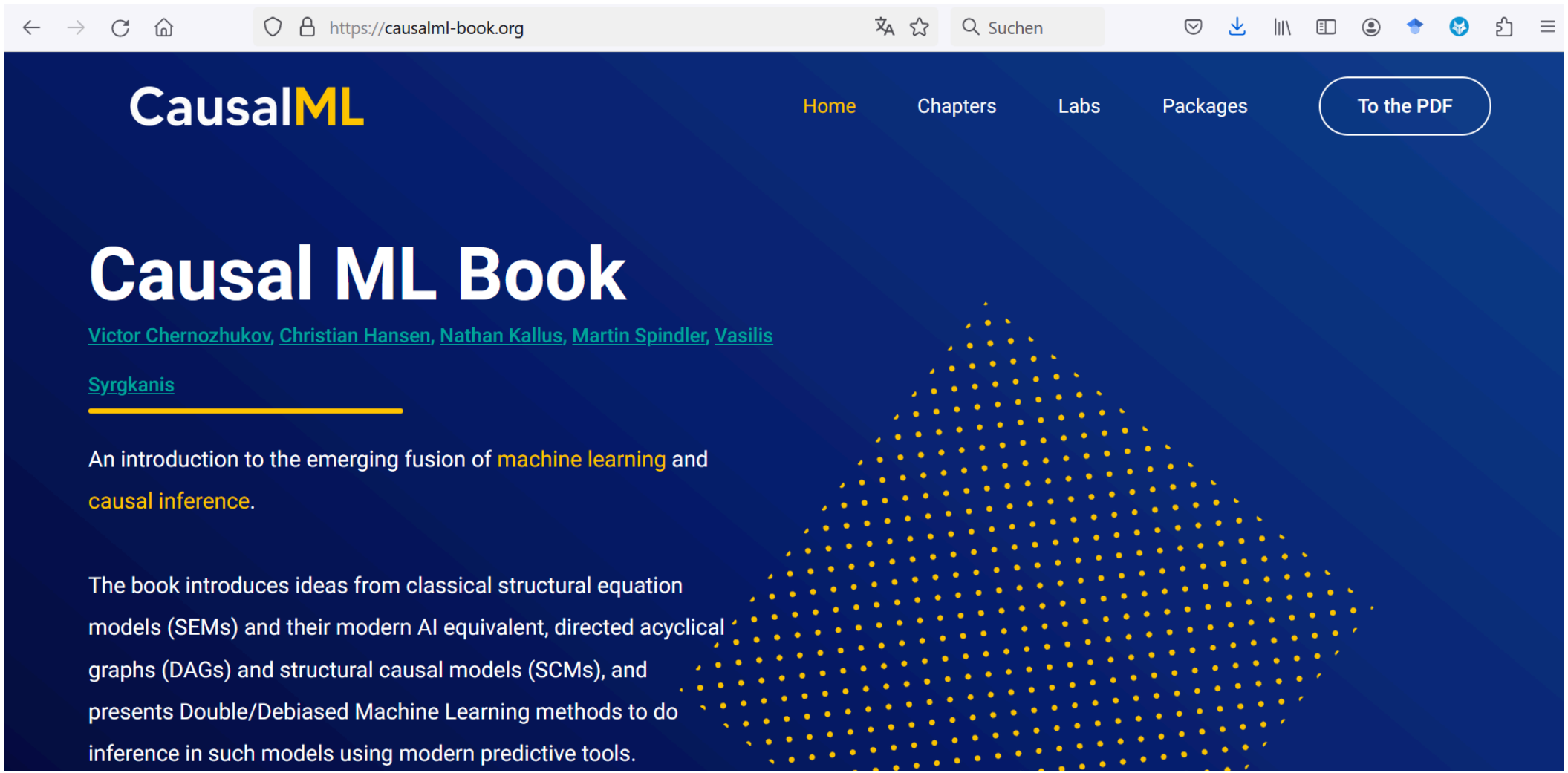
1



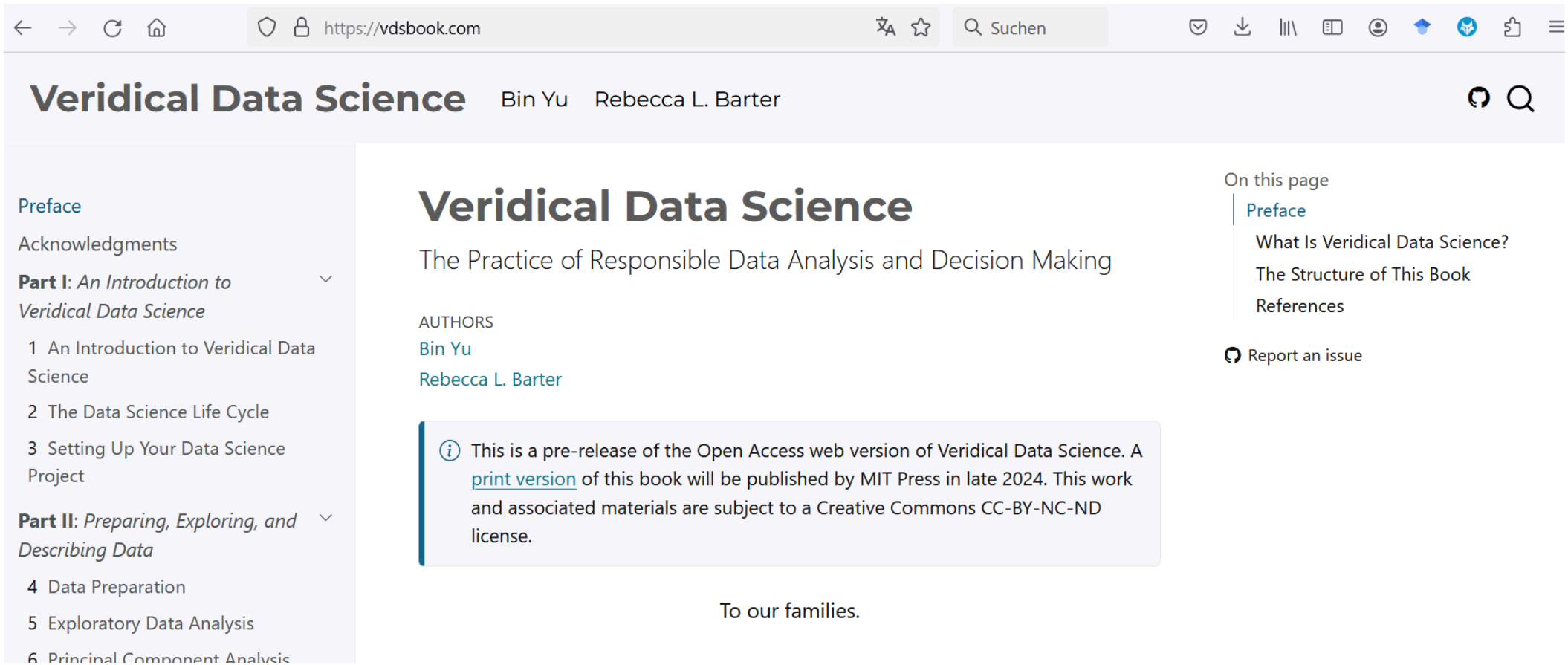
Outline

- Semi-synthetical data for evaluation
- Further applications (Finance, Medicine)
- Extensions of the DoubleML package
- Using multiple time periods and controlling for unobserved confounder

Double Machine Learning



Veridical Data Science



References

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Affiliations



University of Hamburg

https://www.bwl.uni-hamburg.de/en/statistik/allgemein.html

Thanks for your Attention!